

St Catherine's School



2016 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension 1

General Instructions

- Reading Time – 5 minutes
- Working Time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A reference sheet is provided
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations
- Task Weighting – 40%

Total Marks – 70

Section I Pages 3 – 5

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II Pages 6 – 13

60 marks

- Attempt Questions 11 – 14
- Allow about 1 hours and 45 minutes for this section

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Section I

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

- 1 R is a point $(-2, -1)$ and S is a point $(1, 5)$. Find the coordinates of the point X which divides RS externally in the ratio $5 : 2$.

- (A) $(1, 9)$
- (B) $(3, 9)$
- (C) $\left(\frac{9}{7}, \frac{27}{7}\right)$
- (D) $\left(\frac{3}{7}, \frac{23}{7}\right)$

- 2 Find $\int \sin^2 4x \, dx$.

- (A) $\frac{1}{2}\left(x - \frac{\sin 8x}{8}\right) + c$
- (B) $\frac{1}{2}\left(x - \frac{\cos 8x}{8}\right) + c$
- (C) $\frac{1}{2}(x - \sin 8x) + c$
- (D) $x - \frac{\sin 8x}{8} + c$

- 3 Find the remainder when $P(x) = 2x^3 + x^2 - 13x + 6$ is divided by $(x - 1)$.

- (A) 18
- (B) 6
- (C) 4
- (D) -4

- 4 Evaluate $\cos(\tan^{-1} \frac{1}{2})$.
- (A) $\frac{1}{\sqrt{5}}$
(B) $\frac{2}{\sqrt{5}}$
(C) $\frac{1}{2}$
(D) 2
- 5 What is the domain of the function $f(x) = 2 \sin^{-1}(3x - 2)$?
- (A) $-5 \leq x \leq 1$
(B) $-2 \leq x \leq 2$
(C) $\frac{1}{3} \leq x \leq 1$
(D) $\frac{2}{3} \leq x \leq \frac{3}{2}$
- 6 Seven people attend a dinner party. How many ways can they be arranged around a round table if two particular people must sit apart from each other?
- (A) 480
(B) 240
(C) 720
(D) 120
- 7 How many solutions does the equation $\sin 2x = 4\cos x$ have for $0 \leq x \leq 2\pi$.
- (A) 2
(B) 1
(C) 3
(D) 4

- 8 The polynomial equation $P(x) = 2x^3 + x^2 - 13x + 6$ has 3 roots α , β and γ .
Find $\frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma}$.

(A) $\frac{26}{3}$

(B) $\frac{13}{3}$

(C) $\frac{13}{6}$

(D) $\frac{13}{12}$

- 9 If the acute angle between the lines $y = 2x - 3$ and $mx - y - 1 = 0$ is $\frac{\pi}{4}$,
find the value of m .

(A) $-\frac{1}{2}$

(B) -2

(C) $-\frac{1}{3}$

(D) -3

- 10 Evaluate $\int \frac{2}{16+9x^2} dx$.

(A) $\frac{3}{4} \tan^{-1} \frac{3x}{4} + c$

(B) $\frac{1}{12} \tan^{-1} \frac{4x}{3} + c$

(C) $\frac{1}{6} \tan^{-1} \frac{3x}{4} + c$

(D) $\frac{1}{6} \tan^{-1} \frac{4x}{3} + c$

End of Section I

Section II

60 marks

Attempt Questions 11 – 14

Allow about 1 hours and 45 minutes

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) Use the substitution $u = 2x - 1$, to find **3**

$$\int \frac{x}{\sqrt{2x-1}} dx$$

- (b) Consider the letters of the word CALCULATOR.

- (i) How many different arrangements can be made if there are no restrictions? **1**

- (ii) What is the probability that the letter C's are at either ends? **2**

- (c) Isabella guesses at random the answers to each of 10 multiple choice questions. In each question there are 4 options, only one of which is correct.

- (i) Find the probability that Isabella answers exactly 6 of the 10 questions correctly. Give your answer correct to 3 decimal places. **2**

- (ii) Find the probability that Isabella answers at least two questions correctly. Give your answer correct to 3 decimal places. **2**

Question 11 continues on the following page

Question 11 (continued)

- (d) (i) By sketching on the same set of axes the graphs of $y = \cos^{-1} x$ and $y = \frac{\pi}{4} + x$, explain why the equation $\cos^{-1} x - x - \frac{\pi}{4} = 0$ has only one real solution. **2**
- (ii) Taking $x = 0.5$ as the first approximation to the solution of $\cos^{-1} x - x - \frac{\pi}{4} = 0$, use one application of Newton's method to find a better approximation. Give your answer correct to 2 decimal places. **3**

End of Question 11

Question 12 (15 marks) Use a SEPARATE writing booklet.

- (a) A function is defined by $f(x) = 2 - \frac{2}{x+1}$.
- (i) Show that the points of intersection of $f(x)$ and its inverse function $f^{-1}(x)$ are $(0, 0)$ and $(1, 1)$. **2**
- (ii) Sketch the graph of $y = f(x)$ for domain $x \geq -1$. **2**
- Clearly show any equations of asymptotes and intercepts on the coordinate axes.
- (Use at least one third of the page)
- (iii) On the same set of axes, sketch the graph of the inverse function $y = f^{-1}(x)$. **1**
- Clearly show any equations of asymptotes, intercepts on the coordinate axes and points of intersection with $y = f(x)$.
- (iv) Find an expression for $f^{-1}(x)$ in terms of x and clearly state the restriction on its domain. **2**
- (b) The polynomial $P(x)$ is given by $P(x) = x^3 + (k - 1)x^2 + (1 - k)x - 1$ for some real number k .
- (i) Show that $x = 1$ is a root of the equation $P(x) = 0$. **1**
- (ii) Given that $P(x) = (x - 1)(x^2 + kx + 1)$ and $P(x) = 0$ has only one real root, find the possible value(s) of k . **2**

Question 12 continues on the following page

Question 12 (continued)

- (c) A particle is moving in a straight line. At time t seconds it has displacement x metres from a fixed point O on the line, velocity $v \text{ ms}^{-1}$ given by $v^2 = 32 + 8x - 4x^2$ and acceleration $\ddot{x} \text{ ms}^{-2}$.
- (i) Show that the particle is moving in Simple Harmonic Motion. **1**
- (ii) Find the centre and amplitude of the motion. **3**
- (iii) Find the maximum speed of the particle. **1**

End of Question 12

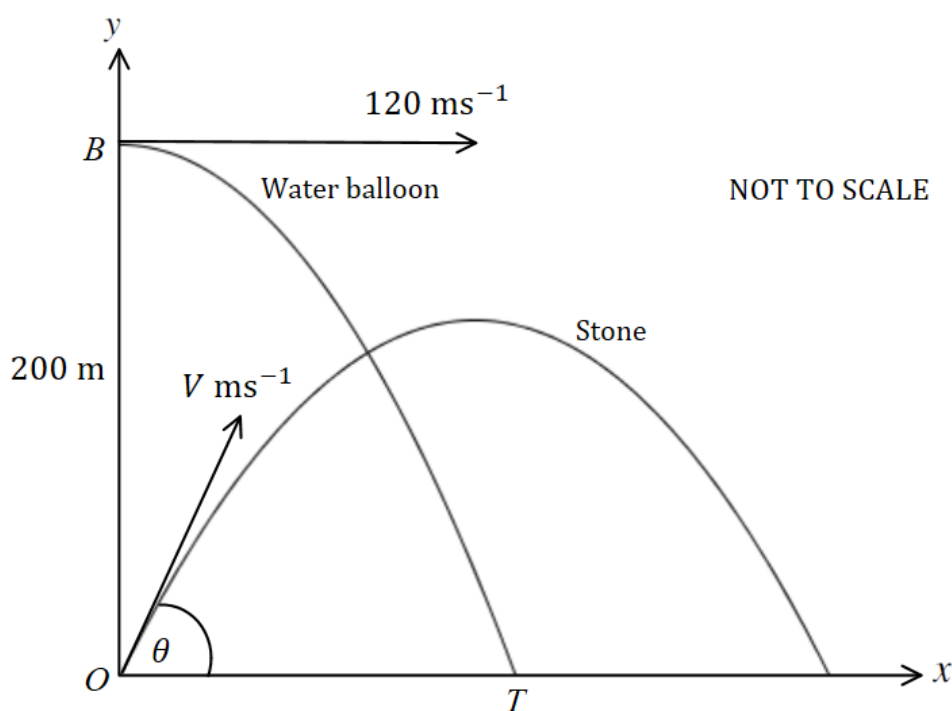
Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) A water balloon is fired horizontally by a cannon from the point B with a velocity of 120 ms^{-1} to reach a target at T.

At the same time, a stone is launched from the point O with a velocity of $V \text{ ms}^{-1}$ and an angle of projection of θ in order to burst the water balloon in the air.

The point O is 200 metres directly below the point B and $\theta = \tan^{-1}\left(\frac{3}{4}\right)$.

Take the acceleration due to gravity as 10 ms^{-2} .



For the water balloon,

- (i) Show that the equations of motion of the water balloon are given by

2

$$x = 120t \quad \text{and} \quad y = -5t^2 + 200.$$

Question 13 continues on the following page

Question 13 (continued)

For the stone, assume that the equations of motion are given by

$$x = Vt \cos \theta \quad \text{and} \quad y = -5t^2 + Vt \sin \theta. \quad (\text{Do NOT prove this.})$$

- (ii) Show that in order for the stone to successfully burst the water balloon in the air, it must be launched at a velocity of 150 ms^{-1} . **2**
- (iii) How high above the ground does the collision occur? **3**
Give your answer correct to the nearest metre.
- (b) Find the exact value of $\sin \left[\cos^{-1} \left(\frac{4}{5} \right) - \tan^{-1} \left(\frac{5}{12} \right) \right]$. **3**
Show all working.
- (c) At time t years the number N of individuals in a population is given by $N = 5000 - 4250e^{-kt}$ for some $k > 0$.
- (i) Find the initial population. **1**
- (ii) Sketch the graph of N as a function of t showing clearly the initial and limiting populations. **2**
- (iii) Find the value of k if $\frac{dN}{dt} = 250$ when N is three times the initial population. **2**

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) Prove by mathematical induction that **3**

$$\sum_{r=1}^n (r^2 + 1)r! = n(n+1)!$$

- (b) In the binomial expansion of $\left(1 - \frac{a}{x}\right)^n$, the coefficient of x^{-4} and the coefficient of x^{-3} are in the ratio of 3 : 2. **3**

Prove that $na - 3a + 6 = 0$.

- (c) Consider the geometric series $1 + (1+x) + (1+x)^2 + \dots + (1+x)^n$, where $x > 0$.

- (i) Show by summation that **1**

$$1 + (1+x) + (1+x)^2 + \dots + (1+x)^n = \frac{(1+x)^{n+1}}{x} - \frac{1}{x}$$

- (ii) Hence, show that **2**

$$\binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \dots + \binom{r}{r} = \binom{n+1}{r+1}$$

Question 14 continues on the following page

Question 14 (continued)

- (d) (i) From a point $A(-p, q)$, where $p > 0$ and $q > 0$, perpendiculars AP and AQ are drawn to meet the x and y axes at $P(-p, 0)$ and $Q(0, q)$ respectively. **2**

Show that the equation of PQ is given by $x = \frac{p}{q}y - p$.

- (ii) Show that the condition for the line PQ to be a tangent to the parabola $y^2 = 4ax$ is $ap - q^2 = 0$. **3**

- (iii) If the points $P(-p, 0)$ and $Q(0, q)$ move on the x and y axes respectively, such that PQ is a tangent to the parabola $y^2 = 4ax$, then the point $A(-p, q)$ traces out a curve. **1**

Find the locus of A .

End of Question 14

End of paper

Student Number: SOLUTION

2016 HIGHER SCHOOL CERTIFICATE
TRIAL EXAMINATION

Mathematics Extension I

Multiple Choice Answer Sheet

Completely fill the response circle representing the most correct answer

	A	B	C	D
1.	☐	☒	☐	☐
2.	☒	☐	☐	☐
3.	☐	☐	☐	☒
4.	☐	☒	☐	☐
5.	☐	☐	☒	☐
6.	☒	☐	☐	☐
7.	☒	☐	☐	☐
8.	☐	☒	☐	☐
9.	☐	☐	☐	☒
10.	☐	☐	☒	☐

2016 Mathematics Extension 1 HSC Trial Examination
SOLUTIONS

Section 1

Question 1 - B

$$R(-2, -1), S(1, 5), m=5, n=-2$$

$$x = \frac{mx_2 + nx_1}{m+n}$$

$$= \frac{5(1) - 2(-2)}{5 + (-2)}$$

$$= \frac{9}{3}$$

$$= 3$$

$$y = \frac{my_2 + ny_1}{m+n}$$

$$= \frac{5(5) - 2(-1)}{5 + (-2)}$$

$$= \frac{27}{3}$$

$$= 9$$

$$\therefore X(3, 9)$$

Question 2 - A

$$\cos 2\theta = 1 - 2\sin^2\theta$$

$$2\sin^2\theta = 1 - \cos 2\theta$$

$$\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\sin^2 4x = \frac{1}{2}(1 - \cos 8x)$$

$$\begin{aligned}\int \sin^2 4x \, dx &= \frac{1}{2} \int 1 - \cos 8x \, dx \\ &= \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + c\end{aligned}$$

Question 3 - D

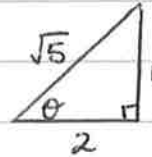
$$\begin{aligned}P(1) &= 2(1)^3 + (1)^2 - 13(1) + 6 \\ &= -4\end{aligned}$$

Question 4 - B

$$\text{let } \theta = \tan^{-1} \frac{1}{2}$$

$$\tan \theta = \frac{1}{2}$$

$$\begin{aligned} \text{then } \cos(\tan^{-1} \frac{1}{2}) &= \cos \theta \\ &= \frac{2}{\sqrt{5}} \end{aligned}$$



Question 5 - C

$$-1 \leq 3x - 2 \leq 1$$

$$1 \leq 3x \leq 3$$

$$\frac{1}{3} \leq x \leq 1$$

Question 6 - A

$$\begin{aligned} \text{No. of ways to sit apart} &= \text{Total ways} - \text{No. of ways to sit together} \\ &= 6! - 5!2! \\ &= 480 \end{aligned}$$

Question 7 - A

$$\sin 2x = 4 \cos x$$

$$2 \sin x \cos x = 4 \cos x$$

$$0 = 2 \sin x \cos x - 4 \cos x$$

$$0 = 2 \cos x (\sin x - 2)$$

$$0 = 2 \cos x$$

$$\text{or } 0 = \sin x - 2$$

$$\cos x = 0$$

$$\sin x = 2$$

$$x = \frac{\pi}{2}, \frac{3\pi}{2}$$

no solution

$$\therefore x = \frac{\pi}{2} \text{ \& } \frac{3\pi}{2} \text{ only i.e. two solutions}$$

Question 8 - B

$$\alpha\beta + \beta\gamma + \alpha\gamma = -\frac{13}{2}$$

$$\begin{aligned}\alpha\beta\gamma &= -\frac{6}{2} \\ &= -3\end{aligned}$$

$$\begin{aligned}\text{then } \frac{2}{\alpha} + \frac{2}{\beta} + \frac{2}{\gamma} &= \frac{2(\beta\gamma + \alpha\gamma + \alpha\beta)}{\alpha\beta\gamma} \\ &= \frac{2(-\frac{13}{2})}{-3} \\ &= \frac{13}{3}\end{aligned}$$

Question 9 - D

$$y = 2x - 3$$

and

$$mx - y - 1 = 0$$

$$m_1 = 2$$

$$y = mx - 1$$

$$m_2 = m$$

$$\tan\theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

$$\tan\frac{\pi}{4} = \left| \frac{2 - m}{1 + 2m} \right|$$

$$1 = \left| \frac{2 - m}{1 + 2m} \right|$$

$$1 = \frac{2 - m}{1 + 2m}$$

or

$$-1 = \frac{2 - m}{1 + 2m}$$

$$1 + 2m = 2 - m$$

$$-1 - 2m = 2 - m$$

$$3m = 1$$

$$-m = 3$$

$$m = \frac{1}{3}$$

$$m = -3 \quad \checkmark$$

Question 10 - c

$$\begin{aligned}\int \frac{2}{16 + 9x^2} dx &= 2 \int \frac{dx}{9\left(\frac{16}{9} + x^2\right)} \\&= \frac{2}{9} \int \frac{dx}{\left(\frac{4}{3}\right)^2 + x^2} \\&= \frac{2}{9} \times \frac{1}{\frac{4}{3}} \tan^{-1}\left(\frac{x}{\frac{4}{3}}\right) + c \\&= \frac{2}{9} \times \frac{3}{4} \tan^{-1}\left(\frac{3x}{4}\right) + c \\&= \frac{1}{6} \tan^{-1}\left(\frac{3x}{4}\right) + c\end{aligned}$$

End of Section 1

Section 2

Question 11

(a) $u = 2x - 1$
 $x = \frac{1}{2}(u + 1)$

$$\frac{du}{dx} = 2$$

$$dx = \frac{1}{2} du$$

$$\begin{aligned} \text{then } \int \frac{x}{\sqrt{2x-1}} dx &= \frac{1}{2} \times \frac{1}{2} \int \frac{u+1}{\sqrt{u}} du \\ &= \frac{1}{4} \int \frac{u}{\sqrt{u}} + \frac{1}{\sqrt{u}} du \\ &= \frac{1}{4} \int u^{\frac{1}{2}} + u^{-\frac{1}{2}} du \\ &= \frac{1}{4} \left(\frac{2}{3} u^{\frac{3}{2}} + 2u^{\frac{1}{2}} \right) + c \\ &= \frac{1}{6} (2x-1)^{\frac{3}{2}} + \frac{1}{2} (2x-1)^{\frac{1}{2}} + c \end{aligned}$$

- 1 mark for correct substitution of dx and x with du and u .
- 2 marks for correct integration in terms of u .
- 3 marks for converting final answer in terms of x & adding constant c .

(b) (i) $\frac{10!}{2!2!2!}$ since there are 2 C's, L's and A's

$$= 453\,600 \text{ arrangements}$$

- 1 mark for correct answer

(ii) C _____ C

Place C's on either ends = 1

Arrange 8 remaining letters = $\frac{8!}{2!2!}$

then no. of ways of arranging C's on either ends = $1 \times \frac{8!}{2!2!}$
= 10080

$$\therefore P(\text{C's on either ends}) = \frac{10080}{453600} \quad \text{using part (i)} \\ = \frac{1}{45}$$

- 1 mark for correct no. of ways of arranging C's and 8 remaining letters
- 2 marks for correctly accounting for repetitions of A and L.

(c) Let X = number of questions answered correctly

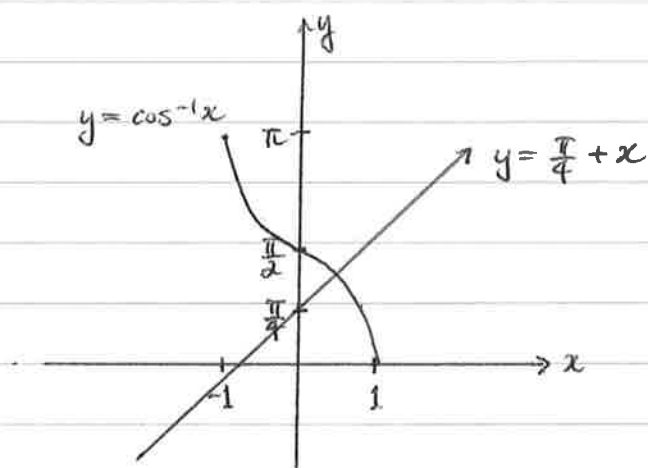
$$(i) \quad P(X=6) = {}^{10}C_6 \left(\frac{1}{4}\right)^6 \left(\frac{3}{4}\right)^4 \\ = 0.016 \quad (3 \text{ decimal places})$$

- 1 mark for showing substantial effort to write correct expression
- 2 marks for correct probability

$$(ii) \quad P(X \geq 2) = 1 - [P(X=1) + P(X=0)] \\ = 1 - \left[{}^{10}C_1 \left(\frac{1}{4}\right) \left(\frac{3}{4}\right)^9 + \left(\frac{3}{4}\right)^{10} \right] \\ = 0.75597... \\ = 0.756 \quad (3 \text{ decimal places})$$

- 1 mark for correct evaluation of probability of $X=0$ or $X=1$
- 2 marks for correct probability

(d) (i)



From the sketch, there is only one intersection point of $y = \cos^{-1}x$ and $y = \frac{\pi}{4} + x$ i.e. $\cos^{-1}x = \frac{\pi}{4} + x$
 $\therefore \cos^{-1}x - \frac{\pi}{4} - x = 0$ has one real solution.

- 1 mark for correctly sketching both equations.
- 2 marks for correct conclusion.

$$(ii) \quad f(x) = \cos^{-1}x - x - \frac{\pi}{4}$$
$$f'(x) = -\frac{1}{\sqrt{1-x^2}} - 1$$

$$\begin{aligned} f(0.5) &= \cos^{-1}0.5 - 0.5 - \frac{\pi}{4} \\ &= \frac{\pi}{3} - 0.5 - \frac{\pi}{4} \\ &= \frac{\pi}{12} - 0.5 \end{aligned}$$

$$\begin{aligned} f'(0.5) &= \frac{-1}{\sqrt{1-0.5^2}} - 1 \\ &= \frac{-1}{\sqrt{\frac{3}{4}}} - 1 \\ &= \frac{-1}{\frac{\sqrt{3}}{2}} - 1 \\ &= -\frac{2}{\sqrt{3}} - 1 \end{aligned}$$

then
$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.5 - \frac{\frac{\pi}{12} - 0.5}{-\frac{2}{\sqrt{3}} - 1}$$

$$= 0.3894 \dots$$

∴ $x = 0.39$ (2 decimal places)

- 1 mark for correct value of $f(0.5)$
- 2 marks for correct differentiation of $f(x)$ & value of $f'(0.5)$
- 3 marks for correct answer

End of Question 11

Question 12

- (a) (i) $f(x)$ & $f^{-1}(x)$ intersect with $y=x$
then find the intersection point of $y=x$ and $y=f(x)$
i.e. $2 - \frac{2}{x+1} = x$

$$2(x+1) - 2 = x(x+1)$$

$$2x + 2 - 2 = x^2 + x$$

$$0 = x^2 - x$$

$$0 = x(x-1)$$

$$\therefore x=0 \quad \text{or} \quad x=1$$

$$\text{and } y=0 \quad \text{or} \quad y=1 \quad \text{since } y=x$$

$\therefore$ intersection points are $(0,0)$ and $(1,1)$

- 1 mark for showing point of intersection occurs for $x=f(x)$.
- 2 marks for clearly solving for two intersection points.

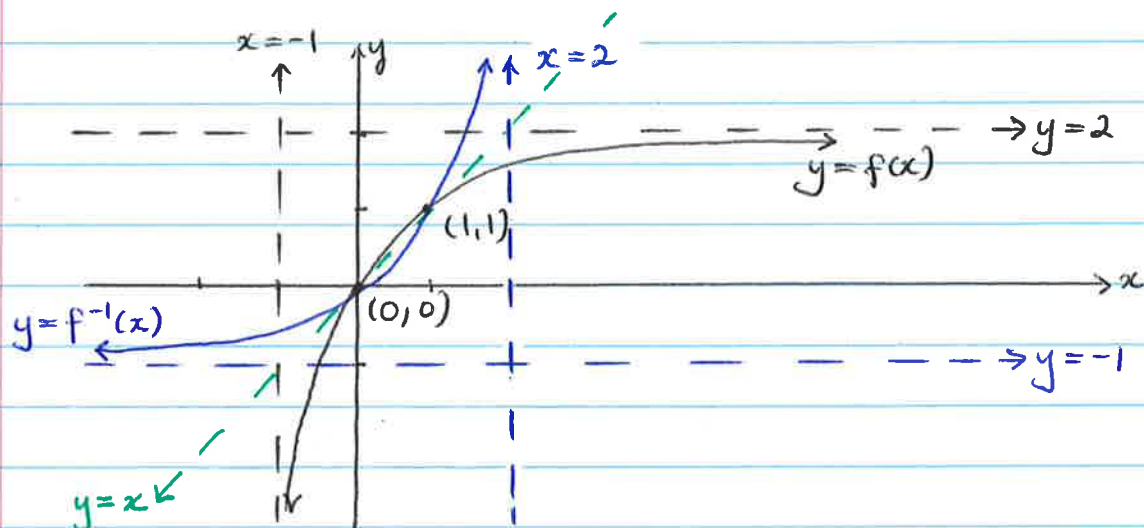
(ii) vertical asymptote : $x=1$

horizontal asymptote : $y=2$

intercept : $(0,0)$

limits : as $x \rightarrow \infty$, $y \rightarrow 2^-$

as $x \rightarrow -1$, $y \rightarrow -\infty$



- 1 mark for correct vertical and horizontal asymptotes
- 2 mark for correct shape of the curve and passing through the intercept $(0,0)$.

(iii) see solution in (ii)

vertical asymptote : $x=2$

horizontal asymptote : $y=-1$

- 1 mark for correct inverse function graph including correct asymptotes, intersection points and shape of the curve.

(iv) $f^{-1}(x) : x = 2 - \frac{2}{y+1}$

$$\frac{2}{y+1} = 2 - x$$

$$y+1 = \frac{2}{2-x}$$

$$\therefore y = \frac{2}{2-x} - 1 \quad \text{where } x < 2$$

$$\text{or } y = \frac{x}{2-x}$$

- 1 mark for swapping x and y variables
- 2 marks for correctly finding the expression for $f^{-1}(x)$.

(b) (i) $P(1) = 1^3 + (k-1) \cdot 1^2 + (1-k) \cdot 1 - 1$
 $= 1 + k - 1 + 1 - k - 1$
 $= 0$

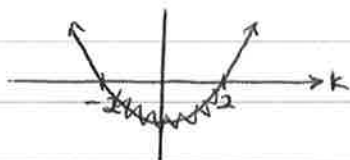
$\therefore$ since $P(1) = 0$, $x=1$ is a root

- 1 mark for correct substitution of $x=1$ into $P(x)$ and showing $P(1) = 0$.

(ii) If $P(x)$ has only one solution and since $x=1$ is a solution, as shown in (i), then $x^2 + kx + 1$ cannot have any real solutions, i.e. $\Delta < 0$ for $x^2 + kx + 1$.

$$\begin{aligned}\text{now } \Delta &= k^2 - 4 \\ &= (k-2)(k+2)\end{aligned}$$

$$\text{then } (k-2)(k+2) < 0$$



$\therefore -2 < k < 2$ for only one solution to exist for $P(x) = 0$.

- 1 mark for showing that one solution exist when $\Delta < 0$ for $x^2 + kx + 1$.
- 2 mark for correct values of k .

$$\begin{aligned}\text{(c) (i) } \ddot{x} &= \frac{d}{dx} \left(\frac{1}{2} v^2 \right) \\ &= \frac{d}{dx} \left[\frac{1}{2} (32 + 8x - 4x^2) \right] \\ &= \frac{d}{dx} (16 + 4x - 2x^2) \\ &= 4 - 4x \\ &= -4(x-1)\end{aligned}$$

$\therefore \ddot{x} = -2^2(x-1)$ is in the form $\ddot{x} = -n^2(x-c)$
hence it's performing a SHM.

- 1 mark for correctly & clearly showing SHM.

(ii) from (i), $n=2$ and $c=1$
 $\therefore$ centre of motion is $x=1$

at endpoints, $v=0$

$$\text{then } 0 = 32 + 8x - 4x^2$$

$$0 = x^2 - 2x - 8$$

$$0 = (x-4)(x+2)$$

then $x=4$ and $x=-2$ are the endpoints of the motion

$$\text{then } a = \frac{4 + 1 - (-2)}{2}$$

$$= 3$$

$\therefore$ amplitude is 3 units

- 1 mark for correct centre of motion
- 2 marks for correctly finding the endpoints of the motion
- 3 marks for correct amplitude.

(iii) maximum speed occurs at centre of motion

$$\text{when } x=1, v^2 = 32 - 8(1) - 4(1)^2$$
$$= 36$$

$\therefore |v| = 6 \text{ ms}^{-1}$ is the max. speed.

- 1 mark for correct maximum speed

End of Question 12

Question 13

(a) (i) $\ddot{x} = 0$ and $\ddot{y} = -10$

$$\dot{x} = \int 0 \, dt$$
$$= c_1$$

at $t=0$, $\dot{x} = V \cos \alpha$

$$= 120 \cos 0$$
$$= 120$$
$$= c_1$$
$$\dot{x} = 120$$
$$x = \int 120 \, dt$$
$$= 120t + c_2$$

at $t=0$, $x=0$

$$= c_2$$

$\therefore x = 120t$

$$\dot{y} = -10t$$
$$y = \int -10t \, dt$$
$$= -5t^2 + c_3$$

at $t=0$, $\dot{y} = V \sin \alpha$

$$= 120 \sin 0$$
$$= 0$$
$$= c_3$$
$$\dot{y} = -10t$$
$$y = \int -10t \, dt$$
$$= -5t^2 + c_4$$

at $t=0$, $y=200$

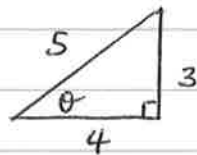
$$= c_4$$

$\therefore y = -5t^2 + 200$

- 1 mark for correctly & clearly deriving vertical equation motion
- 2 marks for correctly & clearly deriving horizontal equation motion

(ii) since $\theta = \tan^{-1}\left(\frac{3}{4}\right)$

$$\tan \theta = \frac{3}{4}$$



then $\cos \theta = \frac{4}{5}$ and $\sin \theta = \frac{3}{5}$

for stone to burst water balloon $x_{WB} = x_s$

i.e. $120t = Vt \cos \theta$

$$120t = Vt \times \frac{4}{5}$$
$$120 = \frac{4}{5}V$$

$\therefore V = 150 \text{ ms}^{-1}$

- 1 mark for correctly showing that stone bursts water balloon when $x_{WB} = x_s$
- 2 marks for correctly and clearly showing that $V = 150 \text{ ms}^{-1}$.

(iii) collision occurs when $y_{WB} = y_s$
i.e. $-5t^2 + 200 = -5t^2 + 120t \sin \theta$

$$200 = 120t \sin \theta$$

$$200 = 120t \times \frac{3}{5} \quad \text{using result from (ii)}$$

$$200 = 90t$$

$$t = \frac{20}{9}$$

$$\text{when } t = \frac{20}{9}, \quad y_{WB} = -5\left(\frac{20}{9}\right)^2 + 200$$

$$= 175.308 \dots$$

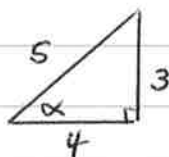
$$= 175 \text{ m (nearest metre)}$$

∴ collision occurs 175 m above the ground.

- 1 mark for showing substantial effort to solve for correct time t .
- 2 marks for correctly solving for time t of collision
- 3 marks for correct height of collision.

(b) let $\alpha = \cos^{-1} \frac{4}{5}$

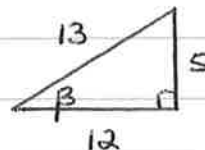
$$\cos \alpha = \frac{4}{5}$$



$$0 < \alpha < \frac{\pi}{2}$$

and $\beta = \tan^{-1} \frac{5}{12}$

$$\tan \beta = \frac{5}{12}$$



$$0 < \beta < \frac{\pi}{2}$$

$$\text{then } \sin(\cos^{-1} \frac{4}{5} - \tan^{-1} \frac{5}{12}) = \sin(\alpha - \beta)$$

$$= \sin \alpha \cos \beta - \sin \beta \cos \alpha$$

$$= \frac{3}{5} \times \frac{12}{13} - \frac{5}{13} \times \frac{4}{5}$$

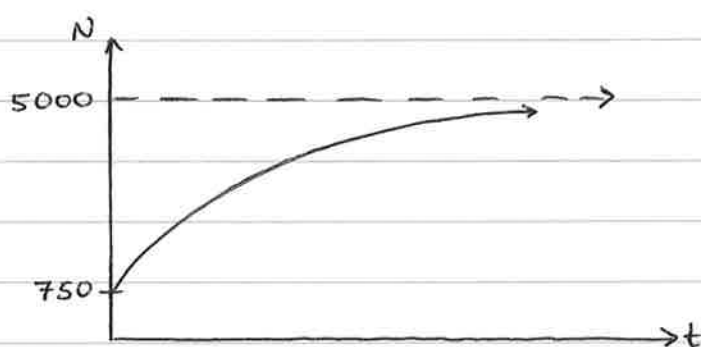
$$= \frac{16}{65}$$

- 1 mark for correct two diagrams representing the given trigonometric ratios.
- 2 marks for correct substitution of $\sin \alpha \cos \beta$
- 3 marks for correct substitution of $\sin \beta \cos \alpha$ and correct answer

(c) (i) when $t=0$, $N = 5000 - 4250e^{-k(0)}$
 $= 5000 - 4250e^0$
 $= 750$

- 1 mark for correct answer

(ii) as $t \rightarrow \infty$, $e^{-kt} \rightarrow 0$
 $4250e^{-kt} \rightarrow 0$
 $5000 - 4250e^{-kt} \rightarrow 5000$



- 1 mark for correct limiting population
- 2 marks for clearly showing the initial population and correct shape of the curve.

(iii) $N = 5000 - 4250e^{-kt}$

$$\frac{dN}{dt} = k \times 4250e^{-kt}$$

$$= k(5000 - N)$$

when $\frac{dN}{dt} = 250$, $N = 3 \times 750$
 $= 2250$

then $250 = k(5000 - 2250)$

$$250 = 2750k$$

$$k = \frac{250}{2750}$$

$$\therefore k = \frac{1}{11}$$

- 1 mark for correctly showing that $\frac{dN}{dt} = k(5000 - N)$
- 2 marks for correctly solving for the value of k .

End of Question 13

Question 14

(a) $\sum_{r=1}^n (r^2+1)r! = n(n+1)!$

i.e. $(1^2+1)1! + (2^2+1)2! + (3^2+1)3! + \dots + (n^2+1)n! = n(n+1)!$

Step 1. Show that the statement is true for $n=1$

$$\text{LHS} = (1^2+1)1!$$

$$= 2 \times 1$$

$$= 2$$

$$\text{RHS} = 1(1+1)!$$

$$= 1 \times 2!$$

$$= 2$$

since $\text{LHS} = \text{RHS}$

the statement is true for $n=1$

Step 2. Assume that the statement is true for $n=k$

i.e. $(1^2+1)1! + (2^2+1)2! + \dots + (k^2+1)k! = k(k+1)!$

Step 3. Show that the statement is true for $n=k+1$

i.e. $(1^2+1)1! + (2^2+1)2! + \dots + (k^2+1)k! + [(k+1)^2+1](k+1)!$
 $= (k+1)(k+2)!$

$$\text{LHS} = k(k+1)! + [(k+1)^2+1](k+1)! \quad \text{using the assumption}$$

$$= (k+1)! [k + (k+1)^2 + 1]$$

$$= (k+1)! (k + k^2 + 2k + 1 + 1)$$

$$= (k+1)! (k^2 + 3k + 2)$$

$$= (k+1)! (k+2)(k+1)$$

$$= (k+2)! (k+1)$$

$$= \text{RHS}$$

$\therefore$ the statement is true for $n=k+1$ if it's true for $n=k$

$\therefore$ the statement is true by mathematical induction

- 1 mark for correctly and clearly showing step 1
- 2 marks for substantial attempt to use the induction assumption with minor errors.
- 3 marks for correctly proving step 3 completely

(b) for $(1 - \frac{a}{x})^n$:

$$\begin{aligned} T_n &= \binom{n}{r} \left(-\frac{a}{x}\right)^r \\ &= \binom{n}{r} (-a)^r x^{-r} \end{aligned}$$

$$\begin{aligned} \text{for } x^{-4}: r=4, \quad T_4 &= \binom{n}{4} (-a)^4 x^{-4} \\ &= \frac{n! a^4}{4! (n-4)!} x^{-4} \end{aligned}$$

$$\begin{aligned} \text{for } x^{-3}: r=3, \quad T_3 &= \binom{n}{3} (-a)^3 x^{-3} \\ &= \frac{-n! a^3}{3! (n-3)!} x^{-3} \end{aligned}$$

$$\text{then } \frac{\text{coefficient of } x^{-4}}{\text{coefficient of } x^{-3}} = \frac{3}{2}$$

$$\frac{\frac{n! a^4}{4! (n-4)!}}{\frac{-n! a^3}{3! (n-3)!}} = \frac{3}{2}$$

$$\frac{a^4 n! 3! (n-3)!}{-a^3 n! 4! (n-4)!} = \frac{3}{2}$$

$$\frac{\cancel{a^4} \cancel{n!} \cancel{3!} (n-3) \cancel{(n-4)!}}{-\cancel{a^3} \cancel{n!} 4 \cdot \cancel{3!} (n-4)!} = \frac{3}{2}$$

$$\frac{-a(n-3)}{4} = \frac{3}{2}$$

$$-2a(n-3) = 12$$

$$-a(n-3) = 6$$

$$-na + 3a = 6$$

$$na - 3a + 6 = 0 \quad \text{as required}$$

- 1 mark for correct expressions of T_4 and T_3
- 2 marks for correctly forming an equation relating the coefficients & its ratio
- 3 marks for clearly & completely deriving the given expression

(c) (i) $a=1$

$$r = 1+x > 0 \quad \text{since } x > 0$$

$$N = n+1$$

$$\text{using } S_N = \frac{a(r^N - 1)}{r - 1}$$

$$= \frac{1 [(1+x)^{n+1} - 1]}{1+x-1}$$

$$= \frac{(1+x)^{n+1} - 1}{x}$$

$$\therefore S_N = \frac{(1+x)^{n+1}}{x} - \frac{1}{x} \quad \text{as required}$$

- 1 mark for clear proof including explicitly stating there are $(n+1)$ terms

(ii) from part (i)

$$1 + (1+x) + (1+x)^2 + \dots + (1+x)^n + \dots + (1+x)^{n-2} + (1+x)^{n-1} + (1+x)^n = \frac{(1+x)^{n+1}}{x} - \frac{1}{x}$$

the coefficient of x^r on LHS is

$$\binom{n}{r} + \dots + \binom{n-2}{r} + \binom{n-1}{r} + \binom{n}{r}$$

the coefficient of x^r on RHS is

$$\begin{aligned} \text{for } \frac{(1+x)^{n+1}}{x} &= \frac{\binom{n+1}{0}x}{x} + \dots + \frac{\binom{n+1}{r}x^r}{x} + \frac{\binom{n+1}{r+1}x^{r+1}}{x} + \dots \\ &= \dots + \binom{n+1}{r}x^{r-1} + \binom{n+1}{r+1}x^r + \dots \end{aligned}$$

so coefficient of x^r is $\binom{n+1}{r+1}$

for $\frac{1}{x}$ there is no term in x^r

$\therefore$ coefficient of x^r on RHS is $\binom{n+1}{r+1}$

then equating coefficients of x^r on LHS and RHS gives:

$$\binom{n}{r} + \binom{n-1}{r} + \binom{n-2}{r} + \dots + \binom{n}{r} = \binom{n+1}{r+1} \quad \text{as required}$$

- 1 mark for clearly deriving the coefficient of x^r on LHS
- 2 marks for clearly deriving the coefficient of x^r on RHS

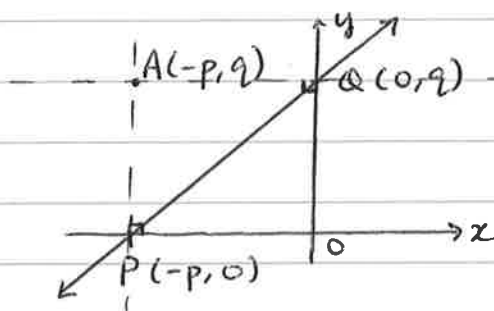
(d) (i) $m = \frac{q}{p}$

y-intercept = q

$$\begin{aligned} \text{then } y &= \frac{q}{p}x + q \\ y - q &= \frac{q}{p}x \end{aligned}$$

$$yp - pq = qx$$

$$\therefore x = \frac{p}{q}y - p \quad \text{as required}$$



- 1 mark for correctly deriving the gradient and y-intercept
- 2 marks for correctly and clearly deriving the equation of the line

(ii) If PQ is a tangent to the parabola then there is only one intersection point of PQ and parabola.

then solving simultaneously for intersection point
 substitute $x = \frac{p}{q}y - p$ into $y^2 = 4ax$

$$y^2 = 4a \left(\frac{p}{q}y - p \right)$$

$$y^2 = \frac{4ap}{q}y - 4ap$$

$$0 = y^2 - \frac{4ap}{q}y + 4ap$$

for one intersection point there is only one solution to the above equation i.e. $\Delta = 0$

$$\text{then } \Delta = \left(\frac{-4ap}{q} \right)^2 - 4(4ap)(1)$$

$$= \frac{16a^2p^2}{q^2} - 16ap$$

$$\text{then } 0 = \frac{16a^2p^2}{q^2} - 16ap$$

$$0 = 16a^2p^2 - 16apq^2$$

$$0 = 16ap(ap - q^2)$$

$$\text{then } 16ap = 0 \quad \text{or} \quad ap - q^2 = 0$$

$$\therefore ap - q^2 = 0 \quad \text{as required.}$$

- 1 mark for correct substitution of the two equations to get a quadratic equation
- 2 marks for correct expression of the discriminant
- 3 marks for correctly and clearly deriving the required expression

(iii) since for A : $x = -p$ and $y = q$
 substitute $p = -x$ & $q = y$ into $ap - q^2 = 0$
 then $-ax - y^2 = 0$
 $\therefore y^2 = -ax$ is the locus of A.

- 1 mark for correct locus of A

End of Question 14

End of Paper